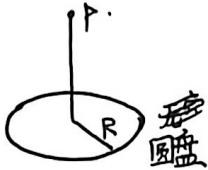


$$j = nqv!$$

P.

实验名称: _____ 姓名: _____ 学号: _____

库仑定理: $F_{12} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r_{12}^2} \hat{e}_{12}$ 电偶极矩: $\vec{p} = q\vec{d}$ $\ominus \xrightarrow{\vec{d}} \oplus$ $\vec{E} = \vec{p} \times \vec{E}$



求P处场强: ①. 求E积分

$$\text{②. } E_z = -\frac{\partial V}{\partial z}$$

$$E_z = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{1+(R/z)^2}}\right)$$

通量 Flux: $\Phi_E = \oint \vec{E} \cdot d\vec{A}$

Gauss' Law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$ (真空, 无介质).

i). 均匀带电壳 $\bigcirc$: $\left\{ \begin{array}{l} \text{外 } E = \frac{\sigma}{4\pi r^2 \epsilon_0} \\ \text{内: } 0 \end{array} \right.$

ii). 带电球体 (实心)

绝缘 $\downarrow$ 与万有引力类似
导体 $\rightarrow$ 电荷仅分布外表面
内部 $E=0$

$$E = \begin{cases} \frac{\sigma r}{4\pi\epsilon_0 R^2} & r < R \\ \frac{\sigma}{4\pi\epsilon_0 r^2} & r > R \end{cases}$$

iii). 均匀带电平面
取高斯面 $\Rightarrow E = \frac{\sigma}{2\epsilon_0}$

电介质 无数个 $\vec{P}$ 组成 $\vec{P}$ 极化位移矢量.

$$\oint \vec{D} \cdot d\vec{A} = \sum q_{\text{free}} \quad \vec{D} = k_e \epsilon_0 \vec{E} = \epsilon_0 \vec{E} + \vec{P}$$

$$\oint \vec{P} \cdot d\vec{A} = \sum q' = -\sum_{\text{in}} q'$$

环路定理 (静电场).

$$\oint \vec{E} \cdot d\vec{l} = 0 \quad \text{感应电场 (变化磁场引发)}$$

$$\oint \vec{E}_{\text{ind}} \cdot d\vec{l} = \epsilon = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_E}{dt} = -\iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A} \quad \text{Stokes公式}$$

$$\oint \nabla \times \vec{E} \cdot d\vec{A} \Rightarrow \text{微分式: } \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

电势能 & 电势

$$\text{电势能 } U(r) = \frac{q_1 q_2}{4\pi\epsilon_0 r} \quad \text{无穷远处电势能为0 } U_{\infty} = 0$$

$$W_{AB} = \int_A^B \vec{F} \cdot d\vec{l} = q_0 \int_A^B \vec{E} \cdot d\vec{l} = -\Delta U = -\int_A^B \vec{E} \cdot d\vec{l}$$

$$\Rightarrow V_P = \int_P^{\infty} \vec{E} \cdot d\vec{l} \quad V_A - V_B = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B}\right)$$

$$\text{球壳 } V(r) = \frac{q}{4\pi\epsilon_0 r}$$

$$r > R \text{ 时: } V(r) = \int_r^{\infty} \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0 r}$$

$$r < R \text{ 时: } V(r) = \int_r^R \vec{E} \cdot d\vec{l} + \int_R^{\infty} \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0 R}$$

静电感应 导体 $\rightarrow$ 导体相连接后 电势相同! 电荷不一定!!



$$\frac{Q_R}{4\pi\epsilon_0 R} = \frac{Q_r}{4\pi\epsilon_0 r} \Rightarrow \frac{Q_R}{Q_r} = \frac{R}{r}$$

$$\sigma = \frac{Q}{4\pi r^2} \quad \frac{\sigma_R}{\sigma_r} = \frac{Q_R}{R^2} \cdot \frac{r^2}{Q_r} = \frac{r}{R}$$

$$W = -\Delta U \Rightarrow r \downarrow \sigma_r \uparrow \text{面电荷密度} \uparrow$$

$$dW = -q_0 dV = \vec{F} \cdot d\vec{l} = q_0 E \cdot dl \cos\theta \Rightarrow E \cos\theta = -\frac{dV}{dl} \Rightarrow E_{\parallel} = -\frac{dV}{dl}$$

$$\vec{E} = -\vec{\nabla} V \quad \vec{\nabla} V = \frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} + \frac{1}{r \sin\theta} \frac{\partial V}{\partial \phi} \hat{\phi}$$

$$\text{球坐标系: } \vec{\nabla} V = \frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} + \frac{1}{r \sin\theta} \frac{\partial V}{\partial \phi} \hat{\phi}$$

电容器: $C = \frac{Q}{V}$

$$C = \frac{\epsilon_0 A}{d} \quad C = \frac{k_e \epsilon_0 A}{d}$$

$\Rightarrow b \rightarrow \infty$ 时: $C = 4\pi\epsilon_0 a$ 一个球也是电容器

$$\text{串联: } \frac{1}{C_{\#}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n} \quad \text{并联: } C_{\#} = C_1 + C_2 + \dots + C_n$$

$$\text{能量 } U = \frac{1}{2} CV^2 = \frac{Q^2}{2C} \Rightarrow dW = V_{(r)} \cdot dq = \frac{Q}{C} \cdot dq \Rightarrow W = \frac{Q^2}{2C}$$

$$\text{能量密度 } u = \frac{U}{V} = \frac{1}{2} \epsilon_0 E^2 \quad \text{普适} \quad \text{一半能量在R, 一半在 } R_0 \Rightarrow R_0 = 2R$$

电质: 电场中的绝缘体 $C = k_e C_0$

$$\oint \vec{D} \cdot d\vec{A} = q_0 \Rightarrow 4\pi r^2 \cdot k_e \epsilon_0 E = q_0 \Rightarrow E = \frac{q_0}{4\pi k_e r^2 \epsilon_0}$$

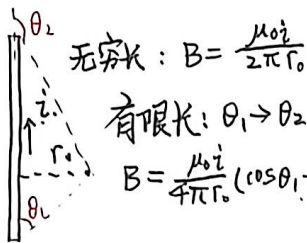
· 磁场

$$d\vec{F}_{12} = \frac{\mu_0}{4\pi} \cdot \frac{i_2 d\vec{s}_2 \times (i_1 d\vec{s}_1 \times \hat{r}_{12})}{r_{12}^2}$$

$$d\vec{F}_{21} = \frac{\mu_0}{4\pi} \times \frac{i_1 d\vec{s}_1 \times (i_2 d\vec{s}_2 \times \hat{r}_{21})}{r_{21}^2}$$

$$\text{定义: } d\vec{F}_2 = i_2 d\vec{s}_2 \times \vec{B}_1 \quad \vec{B}_1 = \frac{\mu_0}{4\pi} \int \frac{i_1 d\vec{s}_1 \times \hat{r}_{12}}{r_{12}^2}$$

$$\text{毕奥-萨伐尔定律: } d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{i d\vec{s} \times \hat{r}}{r^2} \quad \vec{B} = \int \frac{\mu_0}{4\pi} \cdot \frac{i d\vec{s} \times \hat{r}}{r^2}$$



$$\text{无穷长: } B = \frac{\mu_0 i}{2\pi r_0}$$

$$\text{有限长: } \theta_1 \rightarrow \theta_2$$

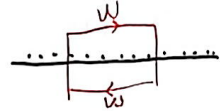
$$B = \frac{\mu_0 i}{4\pi r_0} (\cos\theta_1 - \cos\theta_2)$$



$$B = \frac{\mu_0}{2} \cdot \frac{i R^2}{(R^2 + r^2)^{3/2}} \quad \text{中心处: } B = \frac{\mu_0 i}{2R}$$

螺线管 Solenoid 单位长度 n 匝

Ampere 环路定理解决:

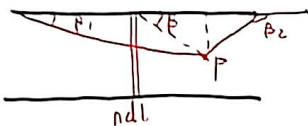


$$\text{无限长} \Rightarrow B = \mu_0 n i$$

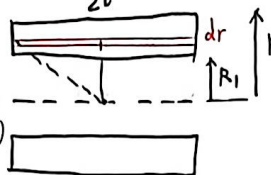
$$\oint \vec{B} \cdot d\vec{l} = 2Bw = \mu_0 i n w$$

$$B = \frac{1}{2} \mu_0 n i (\cos\theta_1 - \cos\theta_2)$$

$$\Rightarrow B = \frac{1}{2} \mu_0 n i \Rightarrow \text{两个加起来 } \frac{1}{2} \mu_0 n i \times 2 = \mu_0 n i$$



推广 多层螺线管, 考虑中心处的 B.



$$\frac{N}{2L(R_2 - R_1)} = j$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

$$\Rightarrow B = \mu_0 j \int_{R_1}^{R_2} \frac{1}{\sqrt{l^2 + r^2}} dr = \mu_0 j \ln \frac{R_2 + \sqrt{R_2^2 + l^2}}{R_1 + \sqrt{R_1^2 + l^2}}$$

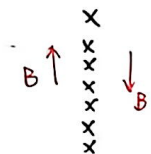
$$B = \frac{1}{2} \mu_0 n i (\cos\theta_1 - \cos\theta_2) = \mu_0 n i \cos\theta_1$$

$$\cos\theta_1 = \frac{l}{\sqrt{l^2 + r^2}}$$

$$n i = \frac{N dr}{2L(R_2 - R_1)} \cdot i \Rightarrow dB = \mu_0 \cdot \frac{N i \cdot \frac{1}{\sqrt{l^2 + r^2}}}{2L(R_2 - R_1)} dr = \frac{\mu_0 N i}{2L(R_2 - R_1)} \cdot \frac{1}{\sqrt{l^2 + r^2}} dr$$

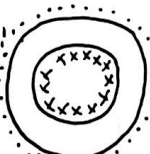
· Magnetic Flux 磁通量 $\oint \vec{B} \cdot d\vec{A} = 0 \text{ T} \cdot \text{m}^2 (\text{Wb})$

· Ampere's Loop Law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 \Sigma i$



$$\oint \vec{B} \cdot d\vec{l} = Bw + 0 + Bw + 0 = 2Bw = \mu_0 n w i \Rightarrow B = \frac{1}{2} \mu_0 n i$$

螺线环:



$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r = \mu_0 n \cdot 2\pi r i \Rightarrow B = \mu_0 n i$$

矩偶极矩 $\vec{p}$, 电偶极矩 $\vec{p}$

$$\vec{p} = \vec{r} \times \vec{B} \quad \vec{p} = \vec{r} \times \vec{E}$$

$$\cdot \text{洛伦兹力 } \vec{F} = q\vec{v} \times \vec{B} \quad F = qvB \sin\theta$$

· 电磁感应定律: $\Phi_B = \int \vec{B} \cdot d\vec{A}$

$$\Rightarrow \mathcal{E} = -\frac{d\Phi_B}{dt}$$

楞次定律

$$\text{① 动生电动势 } \mathcal{E} = \int (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

$\mathcal{E}$: 非静电力将单位正电荷从负极移到正极做的功



$$d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l} = -Bvdr$$

$$\mathcal{E} = -\int_0^l Bvdr = -\frac{1}{2} BvR^2$$

② 感生电动势: (涡旋电场) $\oint \vec{E} \cdot d\vec{l}$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = -\int \frac{\partial \Phi_B}{\partial t} d\vec{l}$$

$$\text{总体有: } \mathcal{E} = \oint (\vec{E} + \vec{v} \times \vec{B}) \cdot d\vec{l}$$

感生 动生

· 原子核磁性:

$$\mu_1 = \frac{e}{2m} \hbar \quad \therefore \vec{\mu}_1 = -\frac{e}{2m} \vec{L}$$

$$L = mvr \text{ 角动量}$$

轨道磁矩

自旋磁矩

$$\mu_B \neq 0 \quad \mu_N \neq 0$$

铁磁材料 温度 ↑ 会向顺磁性

$$\text{转变 } \chi_m = \frac{M}{H} = \frac{M}{T - \theta}$$

Curie-Weiss Law

$\theta > 0$ 是铁磁材料特征参数

$$U = -\int \vec{p} \cdot d\vec{\theta}$$

$$\text{势能 } \vec{p} \cdot \vec{B} \quad \vec{p} \cdot \vec{E}$$

$$U = 0$$

$$\text{正功}$$

$$\text{负功}$$

$$U = \mu B$$

$$U = -\mu B$$

· 互感 (mutual inductance)

$$M_{12} = \frac{\Phi_{12}}{i_2} = \frac{N_2 \Phi_{12}}{i_2} \Rightarrow \mathcal{E}_2 = -\frac{d\Phi_{12}}{dt} = -M_{12} \frac{di_1}{dt}$$

$$\cdot \text{自感 } \Psi = NBA = Li$$

$$\mathcal{E}_1 = -\frac{d\Psi}{dt} = -L \frac{di_1}{dt}$$

$$L = \frac{\Phi_{11}}{i_1}$$

两个线圈 N_1, L_1 与 N_2, L_2

· 无耦合时: $M = \sqrt{L_1 L_2}$

$$\text{② 起在一起时: 同向 } L = L_1 + L_2 + 2\sqrt{L_1 L_2}$$

$$\text{异向 } L = L_1 + L_2 - 2\sqrt{L_1 L_2}$$

磁滞现象与磁畴的转向与排布有关, M 与 H 关系曲线有滞后



$$L = k_m L_0$$

磁导率

电介质

ϵ_0 介电常数

χ_e 极化率

$C = \epsilon_e C_0$

$\vec{P}$ 极化强度矢量

$\vec{D}$ 电位移矢量

$\epsilon_e = 1 + \chi_e$

$P = \chi_e \epsilon_0 E$

$\oint (\epsilon_0 \vec{E} + \vec{P}) \cdot d\vec{A} = \sum q_0$

$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \vec{E} + \chi_e \epsilon_0 \vec{E} = (1 + \chi_e) \epsilon_0 \vec{E} = \epsilon_e \epsilon_0 \vec{E} = \epsilon_e \vec{E}$

ps. $\nabla \cdot \vec{D} = \rho_f$

$\nabla \cdot \vec{P} = -\rho_b$

$L = K_m L_0$ 自感

取向沿 $\vec{E}$

$\vec{P} = \frac{\sum \vec{p}_m}{\Delta V}$

$\vec{P}_m$: 感生电偶极矩

$\vec{P} = \frac{\sum \vec{p}_m}{\Delta V} = n q \vec{l}$ (n : 单位体积分子数)

$\therefore dN = n dV = n l dA \cos \theta$

$dq = q dN = n l q dA \cos \theta$

$= P dA \cos \theta = \vec{P} \cdot d\vec{A}$

$\Rightarrow \oint \vec{P} \cdot d\vec{A} = \sum q'_i = -\sum q''_i$

$\sigma' = \frac{dq}{dA} = P \cos \theta = \vec{P} \cdot \vec{n} = P_n$

$E = E_0 + E'$ $P = \chi_e \epsilon_0 E$

电介质

Gauss' Law: $\oint \vec{E} \cdot d\vec{A} = \frac{\sum q}{\epsilon_0}$

$\Rightarrow \epsilon_0 \oint \vec{E} \cdot d\vec{A} = \sum q$

$\epsilon_0 \oint (\vec{E} + \vec{E}') \cdot d\vec{A} = \sum q_0 + \sum q'_i$

$\Rightarrow \epsilon_0 \oint \vec{E} \cdot d\vec{A} + \oint \vec{P} \cdot d\vec{A} = \sum q_0 - \sum q'_i = -\oint \vec{P} \cdot d\vec{A}$

$\Rightarrow \oint (\epsilon_0 \vec{E} + \vec{P}) \cdot d\vec{A} = \sum q_0$

$\Rightarrow \vec{D} = \epsilon_0 \vec{E} + \vec{P}$

$\oint \vec{D} \cdot d\vec{A} = \sum q_0$ 高斯定理推广

磁介质

K_m 磁导率

χ_m 磁化率

顺磁 同向

抗磁 异向

铁磁 同向

$\vec{M}$ 磁化强度矢量

H 磁场强度

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = \chi_m \mu_0 \vec{H}$

环状分子电流, 组成束缚电流

单位长度密度为 j'

$\Delta V = AL$ $\sum \mu_m = \sum i_m \Delta A = j' LA \Rightarrow j' = \frac{\sum \mu_m}{\Delta V}$

单位体积磁偶极矩 $\vec{M} = \frac{\sum \vec{\mu}_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \vec{B}_0 + \vec{B}_m$ $\oint_L \vec{M} \cdot d\vec{l} = \sum i'$

$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \sum i = \mu_0 \sum (i_0 + i') = \mu_0 \sum i_0 + \mu_0 \sum i' = \mu_0 \sum i_0 + \mu_0 \oint_L \vec{M} \cdot d\vec{l}$

$\Rightarrow \oint_L (\frac{\vec{B}}{\mu_0} - \vec{M}) \cdot d\vec{l} = \sum i_0$ $\oint_L \vec{H} \cdot d\vec{l} = \sum i_0$ 真空中 $\vec{M} = 0$ $\vec{H} = \frac{\vec{B}}{\mu_0}$

$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

$H \cdot \Delta l = N i_0 \Rightarrow H = n i_0$ 单位长度数

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = K_m \mu_0 \vec{H}$

$\vec{M} \times \vec{n} = \vec{j}'$

$\vec{j}' = \frac{\sum \mu_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \mu_0 \vec{H}$

对于圆柱体铁磁介质: $B_0 = \mu_0 n i_0$

$B = K_m \mu_0 H = K_m \mu_0 n i_0 = K_m B_0 \Rightarrow \frac{B}{B_0} = K_m$

补: d

$C_1 = \frac{\epsilon_0 A}{d-b}$ $C_2 = \frac{k_e \epsilon_0 A}{b}$

$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$

顺磁 同向

抗磁 异向

铁磁 同向

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = \chi_m \mu_0 \vec{H}$

环状分子电流, 组成束缚电流

单位长度密度为 j'

$\Delta V = AL$ $\sum \mu_m = \sum i_m \Delta A = j' LA \Rightarrow j' = \frac{\sum \mu_m}{\Delta V}$

单位体积磁偶极矩 $\vec{M} = \frac{\sum \vec{\mu}_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \vec{B}_0 + \vec{B}_m$ $\oint_L \vec{M} \cdot d\vec{l} = \sum i'$

$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \sum i = \mu_0 \sum (i_0 + i') = \mu_0 \sum i_0 + \mu_0 \sum i' = \mu_0 \sum i_0 + \mu_0 \oint_L \vec{M} \cdot d\vec{l}$

$\Rightarrow \oint_L (\frac{\vec{B}}{\mu_0} - \vec{M}) \cdot d\vec{l} = \sum i_0$ $\oint_L \vec{H} \cdot d\vec{l} = \sum i_0$ 真空中 $\vec{M} = 0$ $\vec{H} = \frac{\vec{B}}{\mu_0}$

$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

$H \cdot \Delta l = N i_0 \Rightarrow H = n i_0$ 单位长度数

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = K_m \mu_0 \vec{H}$

$\vec{M} \times \vec{n} = \vec{j}'$

$\vec{j}' = \frac{\sum \mu_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \mu_0 \vec{H}$

顺磁 同向

抗磁 异向

铁磁 同向

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = \chi_m \mu_0 \vec{H}$

环状分子电流, 组成束缚电流

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$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \vec{B}_0 + \vec{B}_m$ $\oint_L \vec{M} \cdot d\vec{l} = \sum i'$

$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \sum i = \mu_0 \sum (i_0 + i') = \mu_0 \sum i_0 + \mu_0 \sum i' = \mu_0 \sum i_0 + \mu_0 \oint_L \vec{M} \cdot d\vec{l}$

$\Rightarrow \oint_L (\frac{\vec{B}}{\mu_0} - \vec{M}) \cdot d\vec{l} = \sum i_0$ $\oint_L \vec{H} \cdot d\vec{l} = \sum i_0$ 真空中 $\vec{M} = 0$ $\vec{H} = \frac{\vec{B}}{\mu_0}$

$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

$H \cdot \Delta l = N i_0 \Rightarrow H = n i_0$ 单位长度数

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = K_m \mu_0 \vec{H}$

$\vec{M} \times \vec{n} = \vec{j}'$

$\vec{j}' = \frac{\sum \mu_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \mu_0 \vec{H}$

顺磁 同向

抗磁 异向

铁磁 同向

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = \chi_m \mu_0 \vec{H}$

环状分子电流, 组成束缚电流

单位长度密度为 j'

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单位体积磁偶极矩 $\vec{M} = \frac{\sum \vec{\mu}_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \vec{B}_0 + \vec{B}_m$ $\oint_L \vec{M} \cdot d\vec{l} = \sum i'$

$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \sum i = \mu_0 \sum (i_0 + i') = \mu_0 \sum i_0 + \mu_0 \sum i' = \mu_0 \sum i_0 + \mu_0 \oint_L \vec{M} \cdot d\vec{l}$

$\Rightarrow \oint_L (\frac{\vec{B}}{\mu_0} - \vec{M}) \cdot d\vec{l} = \sum i_0$ $\oint_L \vec{H} \cdot d\vec{l} = \sum i_0$ 真空中 $\vec{M} = 0$ $\vec{H} = \frac{\vec{B}}{\mu_0}$

$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

$H \cdot \Delta l = N i_0 \Rightarrow H = n i_0$ 单位长度数

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = K_m \mu_0 \vec{H}$

$\vec{M} \times \vec{n} = \vec{j}'$

$\vec{j}' = \frac{\sum \mu_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \mu_0 \vec{H}$

顺磁 同向

抗磁 异向

铁磁 同向

$\vec{M} = \chi_m \vec{H}$

$\vec{B} = \chi_m \mu_0 \vec{H}$

环状分子电流, 组成束缚电流

单位长度密度为 j'

$\Delta V = AL$ $\sum \mu_m = \sum i_m \Delta A = j' LA \Rightarrow j' = \frac{\sum \mu_m}{\Delta V}$

单位体积磁偶极矩 $\vec{M} = \frac{\sum \vec{\mu}_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \vec{B}_0 + \vec{B}_m$ $\oint_L \vec{M} \cdot d\vec{l} = \sum i'$

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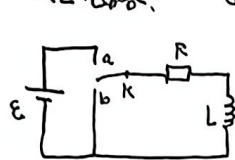
$\vec{M} \times \vec{n} = \vec{j}'$

$\vec{j}' = \frac{\sum \mu_m}{\Delta V}$

$\oint \vec{M} \cdot d\vec{l} = \sum i'$ 束缚电流

$\vec{B} = \mu_0 \vec{H}$

RL电路 ①. 充电 a



$$iR + L \frac{di}{dt} = \varepsilon$$

$$\frac{di}{dt} = \frac{1}{L}(\varepsilon - iR) = -\frac{R}{L}(i - \frac{\varepsilon}{R}) \Rightarrow i - \frac{\varepsilon}{R} = C' e^{-\frac{R}{L}t} \quad C' = -\frac{\varepsilon}{R}$$

$$\Rightarrow i = \frac{\varepsilon}{R}(1 - e^{-Rt/L}). \quad L \text{ 上电压: } V_L = L \frac{di}{dt} = \varepsilon e^{-Rt/L}.$$

②. 放电 b. $iR + L \frac{di}{dt} = 0 \Rightarrow i = \frac{\varepsilon}{R} e^{-Rt/L} \quad V_L = L \frac{di}{dt} = -\varepsilon e^{-Rt/L}$

能量: eg. Capacitor: $U_e = \frac{Q^2}{2C} = \frac{1}{2} CV^2 \quad u_E = \frac{1}{2} \varepsilon_0 E^2$

L 中: $dW = -\varepsilon_L dq = -\varepsilon_L i dt \Rightarrow dW = Li di \Rightarrow W = \frac{1}{2} Li_{max}^2 = \frac{1}{2} LI^2$

能量密度: $u_B = \frac{U}{V} = \frac{\frac{1}{2} LI^2}{\lambda A} = \frac{\frac{1}{2} \mu_0 n^2 \lambda I^2}{\lambda A} = \frac{1}{2} \mu_0 n^2 I^2 = \frac{B^2}{2\mu_0} \quad u_e = \frac{1}{2} \varepsilon_0 E^2.$

Vacuum: $u_B = \frac{B^2}{2\mu_0} = \frac{1}{2} \vec{B} \cdot \vec{H} \quad u_E = \frac{1}{2} \varepsilon_0 E^2 = \frac{1}{2} \vec{D} \cdot \vec{E}$

$\Rightarrow$ 两个线圈 $\begin{matrix} N_1 L_1 \\ N_2 L_2 \end{matrix}$ 有互感 $\begin{matrix} M_{12} \\ M_{21} \end{matrix}$

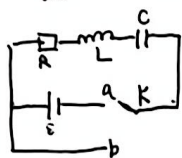
互感能量 $W = W_1 + W_2 = -\int_0^\infty \varepsilon_{21} i_1 dt - \int_0^\infty \varepsilon_{12} i_2 dt$

$= M I_1 I_2 = \frac{1}{2} M_{12} I_1 I_2 + \frac{1}{2} M_{21} I_1 I_2$

$U = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2$

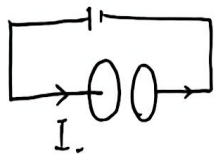
电磁振荡: $U = U_B + U_E = \frac{1}{2} Li^2 + \frac{1}{2} \frac{q^2}{C} \Rightarrow \frac{dU}{dt} = Li \frac{di}{dt} + \frac{q}{C} \frac{dq}{dt} = Li \frac{d^2 q}{dt^2} + \frac{q}{C} \dot{q} = 0 \Rightarrow \frac{d^2 q}{dt^2} + \frac{1}{LC} q = 0 \Rightarrow \omega = \sqrt{\frac{1}{LC}}.$

$$L \frac{di}{dt} + iR + \frac{q}{C} = \begin{cases} \varepsilon \\ 0 \end{cases}$$



阻尼 $\begin{cases} \text{过阻尼} \\ \text{临界} \\ \text{欠阻尼} \end{cases}$

位移电流: $\oint \vec{H} \cdot d\vec{l} = i_0 + i_D.$



平行板电容器.

$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A}.$

$\oint \vec{D} \cdot d\vec{A} = q_0 \quad \oint \vec{j}_0 \cdot d\vec{A} = -\frac{dq_0}{dt} = -\oint \frac{\partial \vec{D}}{\partial t} \cdot d\vec{A} \Rightarrow \oint (\vec{j}_0 + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{A} = 0 \quad q = \varepsilon_0 A E$

新安培环路定理: $\oint \vec{H} \cdot d\vec{l} = i_0 + i_D = \iint (\vec{j}_0 + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{A} \quad i_0 = \frac{dq_0}{dt} = \varepsilon_0 \frac{d\Phi_E}{dt} = -\frac{d\Phi_D}{dt} = i_D.$

麦克斯韦方程组:

$$\begin{cases} \oint \vec{D} \cdot d\vec{A} = q_0 \\ \oint \vec{B} \cdot d\vec{A} = 0 \\ \oint \vec{E} \cdot d\vec{l} = -\iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A} \\ \oint \vec{H} \cdot d\vec{l} = i_0 + \iint \frac{\partial \vec{D}}{\partial t} \cdot d\vec{A} \end{cases} \rightarrow \begin{cases} \nabla \cdot \vec{D} = \rho_{e0} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} = \vec{j}_0 + \frac{\partial \vec{D}}{\partial t} \end{cases}$$

$\rho_{e0} = 0, \vec{j}_0 = 0$ 真空中: $\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \rightarrow \nabla \cdot \vec{H} = 0 \\ \nabla \times \vec{E} = -k_m \mu_0 \frac{\partial \vec{H}}{\partial t} \\ \nabla \times \vec{H} = k_e \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \end{cases}$

$$\Rightarrow \begin{cases} \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0 \\ \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} = 0 \end{cases}$$

用各.

$$\Rightarrow \begin{cases} \frac{\partial^2 E_x}{\partial z^2} - k_e \varepsilon_0 k_m \mu_0 \frac{\partial^2 E_x}{\partial t^2} = 0 \\ \frac{\partial^2 H_y}{\partial z^2} - k_e \varepsilon_0 k_m \mu_0 \frac{\partial^2 H_y}{\partial t^2} = 0 \end{cases} \Rightarrow v = \frac{1}{\sqrt{k_e \varepsilon_0 k_m \mu_0}}$$

$$\sqrt{k_e \varepsilon_0} E_0 = \sqrt{k_m \mu_0} H_0.$$

真空中: $k_m = k_e = 1 \Rightarrow \sqrt{\varepsilon_0} E_0 = \sqrt{\mu_0} H_0.$

$$\Rightarrow B_0 = \frac{E_0}{c} \quad c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}.$$

实验名称: _____ 姓名: _____ 学号: _____

坡印廷矢量 $\vec{S} = \vec{E} \times \vec{H}$ (单位时间内通过单位面积的电磁能量)

电磁波能量: $U = \iiint (\frac{1}{2} \epsilon_0 E^2 + \frac{B^2}{2\mu_0}) dV = \iiint (\frac{1}{2} \vec{E} \cdot \vec{D} + \frac{1}{2} \vec{B} \cdot \vec{H}) dV \Rightarrow \frac{dU}{dt} = -\oint \vec{S} \cdot d\vec{A} - Q + P$

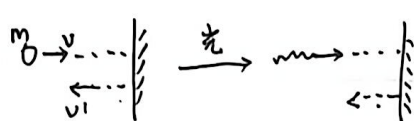
电磁波中 $U_E = U_B$ $[U_E = \frac{1}{2} \epsilon_0 E^2; U_B = \frac{B^2}{2\mu_0} \quad \sqrt{\epsilon_0 \mu_0} H = \sqrt{\epsilon_0 \mu_0} E \quad \text{真空中} \Rightarrow \sqrt{\mu_0} H = \sqrt{\epsilon_0} E]$

$$U_E = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 c^2 B^2 = \frac{1}{2} \epsilon_0 \frac{1}{\epsilon_0 \mu_0} B^2 = \frac{B^2}{2\mu_0} = U_B$$

$$\therefore U = U_B + U_E = \epsilon_0 E^2$$

平均能量密度 $\langle u \rangle = \epsilon_0 \langle E^2 \rangle = \epsilon_0 E_{rms}^2 \langle \sin^2(\omega t - kx) \rangle = \frac{\epsilon_0 E_{max}^2}{2} \quad \langle u \rangle = \epsilon_0 E_{rms}^2 = \epsilon_0 (\frac{E_{max}}{\sqrt{2}})^2$

波的强度: $I = c \langle u \rangle = c \cdot \frac{\epsilon_0 E_{max}^2}{2}$ (单位时间内通过单位面积的电磁能)



$$\Delta \vec{F} \cdot c \Delta t = (\vec{S}_{in} - \vec{S}_{ref}) \Delta A \cdot \Delta t$$

$$\therefore \Delta \vec{F} = \frac{1}{c} (\vec{S}_{in} - \vec{S}_{ref}) \Delta A$$

光压: $P = \frac{|\Delta \vec{F}|}{\Delta A} = \frac{1}{c} (|\vec{S}_{in}| + |\vec{S}_{ref}|)$

动量: $\Delta \vec{G}_p = \Delta \vec{F} \cdot \Delta t = \frac{1}{c} (\vec{S}_{in} - \vec{S}_{ref}) \Delta A \cdot \Delta t$

动量密度: $\frac{\Delta \vec{G}_p}{\Delta V} = \frac{\frac{1}{c} (\vec{S}_{in} - \vec{S}_{ref}) \Delta A \Delta t}{\Delta A \cdot c \cdot \Delta t} = \frac{1}{c^2} (\vec{S}_{in} - \vec{S}_{ref}) = \Delta \vec{g}$

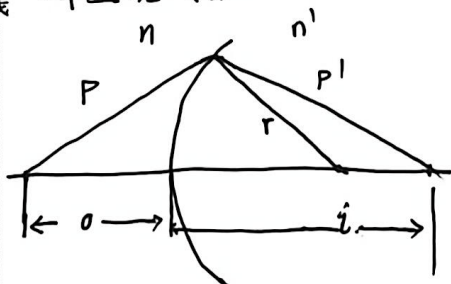
全反射

$$P = \frac{2S}{c} = \frac{2}{c} EH$$

$$\begin{cases} \vec{g}_{in} = \frac{\vec{S}_{in}}{c^2} \\ \vec{g}_{out} = \frac{\vec{S}_{ref}}{c^2} \end{cases}$$

订

线 球面镜成像:



$$\frac{n}{o} + \frac{n'}{i} = \frac{n' - n}{r}$$

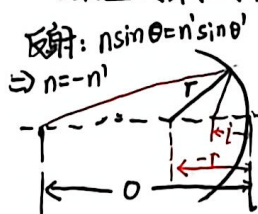
$i \rightarrow \infty$ 时, 第一焦点: $o = f = \frac{nr}{n' - n}$

$o \rightarrow \infty$ 时, 第二焦点: $i = f' = \frac{n'r}{n' - n}$

$$\frac{f}{f'} = \frac{n}{n'} \quad \frac{f}{o} + \frac{f'}{i} = 1$$

符号: $o > 0$ 实物
 $o < 0$ 虚物
 $i > 0$ 实像
 $i < 0$ 虚像
 $r > 0$ 凸
 $r < 0$ 凹

球面反射成像 ($n = -n'$)



反射: $n \sin \theta = n' \sin \theta'$

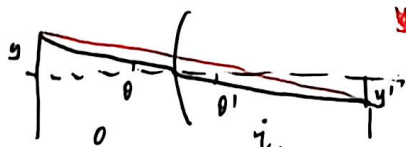
$$\Rightarrow n = -n'$$

由 $\frac{n}{o} + \frac{n'}{i} = \frac{n' - n}{r}$; $\frac{f}{o} + \frac{f'}{i} = 1$

$$\Rightarrow \frac{n}{o} + \frac{-n}{(-i)} = \frac{-2n}{r} \Rightarrow \frac{f}{o} + \frac{f'}{i} = -\frac{2}{r}$$

横向放大率: $m = \frac{y'}{y} = -\frac{i \theta'}{o \theta} = -\frac{n i}{n' o}$

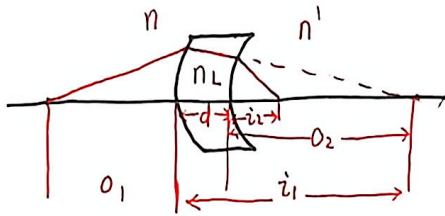
$\Rightarrow$ 对反射成像: $m = \frac{i}{o}$



$$n \theta \approx n' \theta'$$

$$\begin{aligned} y &= o \theta \\ y' &= i \theta' \end{aligned} \Rightarrow \frac{y'}{y} = -\frac{i \theta'}{o \theta} = -\frac{i}{o} \frac{n}{n'}$$

★ 磨镜者公式 (薄透镜).



$$o_2 = -(i_1 - d) \quad d \approx 0$$

$$\Rightarrow o_2 = -i_1$$

$$\Rightarrow \begin{cases} \frac{f_1}{o_1} + \frac{f_1'}{i_1} = 1 \longrightarrow \frac{f_1 f_2}{o_1} + \frac{f_1' f_2}{i_1} = f_2 \\ \frac{f_2}{-i_1} + \frac{f_2'}{i_2} = 1 \longrightarrow \frac{f_1' f_2}{-i_1} + \frac{f_1' f_2'}{i_2} = f_1' \end{cases}$$

$$\text{相加} \Rightarrow \frac{f_1 f_2}{o_1} + \frac{f_1' f_2'}{i_2} = f_1' + f_2$$

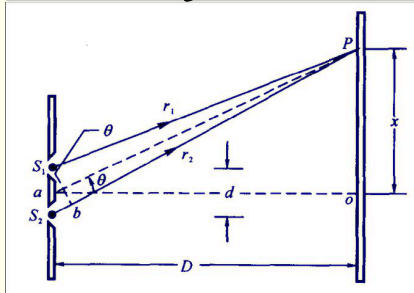
$$\Rightarrow \text{整个过程: } \frac{f_1 f_2}{o} + \frac{f_1' f_2'}{i} = f_1' + f_2 \Rightarrow \begin{cases} f = \frac{f_1 f_2}{f_1' + f_2} \\ f' = \frac{f_1' f_2'}{f_1' + f_2} \end{cases}$$

$$f = \frac{\frac{n n_L r_1 r_2}{(n_L - n)(n' - n_L)}}{\frac{n_L r_1}{n_L - n} + \frac{n_L r_2}{n' - n_L}} = \frac{n n_L r_1 r_2}{(n' - n_L) n_L r_1 + (n_L - n) n_L r_2} = \frac{n}{\frac{(n' - n_L)}{r_2} + \frac{(n_L - n)}{r_1}}$$

$$f' = \frac{n'}{\frac{n_L - n}{r_1} + \frac{n' - n_L}{r_2}} \Rightarrow \frac{f'}{f} = \frac{n'}{n} \quad n = n' = 1 \text{ 时}$$

$$\Rightarrow f = f' = \frac{1}{(n_L - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right)}$$

9. 光的干涉



极大: $\delta = d \sin \theta = k\lambda$
 极小: $\delta = d \sin \theta = (k + \frac{1}{2})\lambda$ $k \in \mathbb{Z}$

$\Delta \varphi = 2\pi \cdot \frac{\delta}{\lambda} \approx \frac{2\pi}{\lambda} \cdot d \sin \theta$ 观察屏上位置: $\begin{cases} x = \frac{D}{d} k \lambda \\ x = \frac{D}{d} (k + \frac{1}{2}) \lambda \end{cases} k \in \mathbb{Z}$

相干光波: 频率相同、振动方向相同、相位差恒定

相邻明/暗纹: $\Delta x = \frac{D}{d} \lambda$

$\begin{cases} E_1 = E_{10} \cos(\omega t + \varphi_1) \\ E_2 = E_{20} \cos(\omega t + \varphi_2) \end{cases} \Rightarrow \text{合成 } \vec{E} = \vec{E}_1 + \vec{E}_2 = E_0 \cos(\omega t + \varphi)$
 $E_0 \cos \varphi = E_{10} \cos \varphi_1 + E_{20} \cos \varphi_2$ $E_0 \sin \varphi = E_{10} \sin \varphi_1 + E_{20} \sin \varphi_2$
 $\varphi = \arctg \frac{E_{10} \sin \varphi_1 + E_{20} \sin \varphi_2}{E_{10} \cos \varphi_1 + E_{20} \cos \varphi_2}$ $E_0^2 = E_{10}^2 + E_{20}^2 + 2E_{10}E_{20} \cos(\varphi_2 - \varphi_1)$

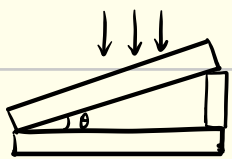
光强: $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\varphi_2 - \varphi_1)$ 中央附近 $I_1 \approx I_2 = I_0$ $I = 2I_0 [1 + \cos(\varphi_2 - \varphi_1)] = 4I_0 \cos^2(\frac{\varphi_2 - \varphi_1}{2})$

薄膜干涉:

① $\lambda_1 = \frac{v}{f} = \frac{c}{n_1 f} = \frac{\lambda}{n_1}$
 $\frac{2d}{\lambda_1} = \frac{2d \cdot n_1}{\lambda}$ 再判断是否半波损失与 $\begin{cases} = k \\ = k + \frac{1}{2} \end{cases}$
 ② 光程: $n r$ 光程差 $\delta = n_2 r_2 - n_1 r_1$

相位差取决于光程差
 而非几何路程差
 $\therefore \Delta \varphi = \frac{2\pi}{\lambda} \cdot \delta = \frac{2\pi}{\lambda} (n_1 \cdot 2d - 0) \Rightarrow \frac{\Delta \varphi}{2\pi} = \frac{n_1 \cdot 2d}{\lambda}$

劈尖干涉:

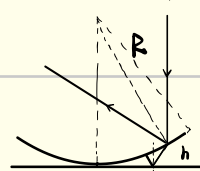


h 时, $\delta = 2nh = k\lambda$

$h + \Delta h$ 时, $\delta = 2n(h + \Delta h) = (k + 1)\lambda$ 下一级条纹

空气中 $n = 1$

$\therefore 2n\Delta h = \lambda$ $\Delta h = \frac{\lambda}{2n} \Rightarrow \tan \theta = \frac{\Delta h}{\Delta x} \approx \theta \therefore \theta \approx \frac{\lambda}{2n\Delta x} = \frac{\lambda}{2\Delta x}$



牛顿环:

有半波损失 $2nh + \frac{\lambda}{2} = \begin{cases} k\lambda, \text{明纹} \\ (k + \frac{1}{2})\lambda, \text{暗纹} \end{cases} k \in \mathbb{Z}$

$n = 1$; 明半径 $r = \sqrt{(k - \frac{1}{2})R\lambda}$

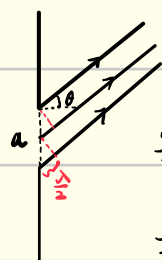
$k \in \mathbb{Z}$

暗半径: $r = \sqrt{k\lambda R}$

衍射与光栅

衍射: 遇障碍物偏离直线传播

Fraunhofer 单缝衍射



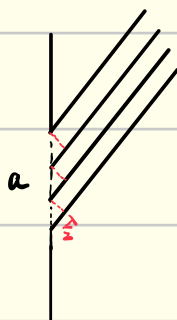
偶数个半波带

无数对这样的两条光线均干涉

相消 $\therefore \delta = a \sin \theta = 2k \cdot \frac{\lambda}{2}$

$$\Rightarrow a \sin \theta = k \lambda$$

暗



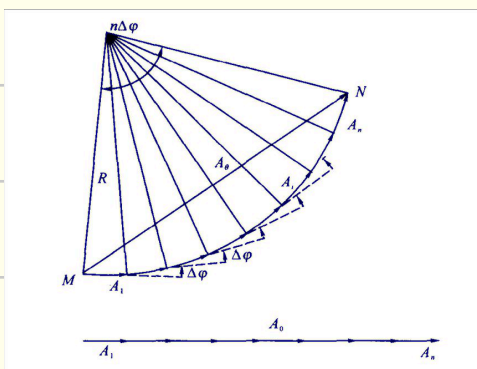
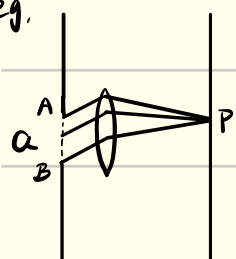
奇数个半波带, 上部分两两成对

相消, 下方始终还有一个半波带, $\Rightarrow$ 亮纹

$$\delta = a \sin \theta = (2k+1) \cdot \frac{\lambda}{2} \Rightarrow a \sin \theta = (k+\frac{1}{2}) \lambda$$

矢量图解: 由于在a处光是连续的, 因此无数条光到P的

eg. 相位也是连续的从A到B



$$\widehat{MN} = R(n\Delta\varphi) = nA_1$$

$$\text{合振幅 } A_\theta = 2R \sin \frac{n\Delta\varphi}{2} = 2 \frac{nA_1}{n\Delta\varphi} \sin \frac{n\Delta\varphi}{2}$$

$$n\Delta\varphi = \frac{2\pi}{\lambda} a \sin \theta \quad (\text{总光程差})$$

$$\therefore A_\theta = nA_1 \frac{\sin(\frac{\pi a}{\lambda} \sin \theta)}{\frac{\pi a \sin \theta}{\lambda}} = nA_1 \frac{\sin u}{u}$$

展成直线为A0
中央明纹中心
处的振幅
 $\theta=0$

光强与振幅平方成正比!

$$\Rightarrow A_\theta = A_0 \frac{\sin u}{u}$$

$$\text{光强: } \frac{I}{I_0} = \frac{A_\theta^2}{A_0^2} = \frac{\sin^2 u}{u^2}$$

$$\therefore I = I_0 \frac{\sin^2 u}{u^2} \quad (u = \frac{\pi a}{\lambda} \sin \theta)$$

$$u = \frac{n\Delta\varphi}{2} = \frac{2\pi}{\lambda} \cdot a \sin \theta$$

当 $\frac{\pi a}{\lambda} \sin \theta = k\pi$, $k \in \mathbb{Z}$ 时, I 最小

当 $\frac{\pi a}{\lambda} \sin \theta \rightarrow 0$ 时, $I = I_0$ 最大. ($\theta=0$)

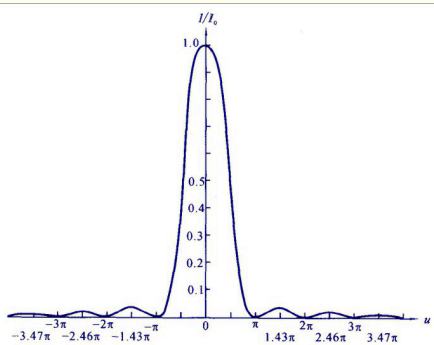
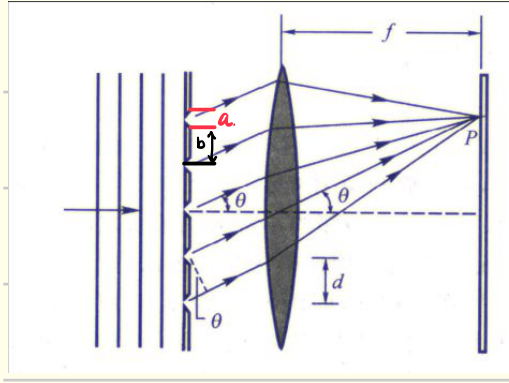


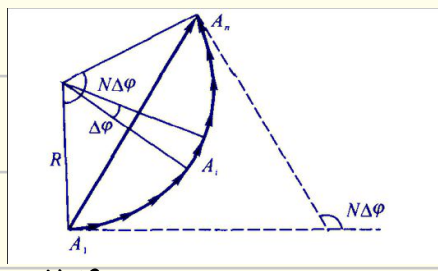
图 17.14 单缝夫琅禾费衍射相对光强分布

光栅衍射



相邻两缝

$d = a + b$, 共 N 条缝, 相邻两缝相位差: $\Delta\varphi = \frac{2\pi}{\lambda} d \sin\theta$



总相位差: $N\Delta\varphi$

$$A = 2R \sin \frac{N\Delta\varphi}{2}$$

$$u = \frac{\pi a \sin\theta}{\lambda}$$

$$A_i = 2R \sin \frac{\Delta\varphi}{2} \quad A_i = A_0 \frac{\sin u}{u}$$

$$\Rightarrow A = \frac{A_i \sin \frac{N\Delta\varphi}{2}}{\sin \frac{\Delta\varphi}{2}} = A_0 \frac{\sin u}{u} \cdot \frac{\sin \frac{N\Delta\varphi}{2}}{\sin \frac{\Delta\varphi}{2}}$$

$$\Rightarrow I = I_0 \left(\frac{\sin u}{u} \right)^2 \cdot \left(\frac{\sin \frac{N\Delta\varphi}{2}}{\sin \frac{\Delta\varphi}{2}} \right)^2$$

$$u = \frac{\pi a \sin\theta}{\lambda} \quad \frac{\Delta\varphi}{2} = \frac{\pi d \sin\theta}{\lambda}$$

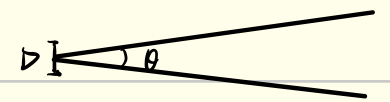
单缝

缝之间

光栅的分辨本领: 把波长相近的两条谱线分辨清楚

$$R = \frac{\lambda}{\Delta\lambda} \xrightarrow{\text{平均波长}} \xrightarrow{\text{波长差}} = KN \quad (\text{衍射级次 } k)$$

$$d \sin\theta = k(\lambda + \Delta\lambda)$$



$$Nd \sin\theta = k'\lambda = (KN+1)\lambda$$

圆孔衍射——最小分辨角 $\theta_{min} = 1.22 \frac{\lambda}{D}$

$$\text{分辨本领: } R = \frac{1}{\theta_{min}}$$

偏振

线偏振光: 传播过程中, 空间各点光矢量都沿同一个固定方向振动.

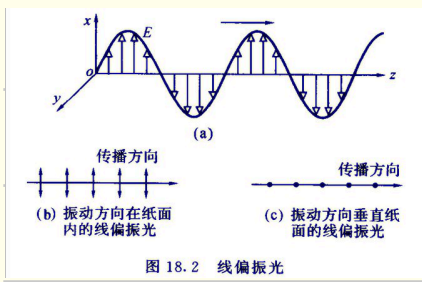


图 18.2 线偏振光

- Right(右)-handed (RCP)
- Electric field spirals CW in space (at fixed time)

RCP = CW

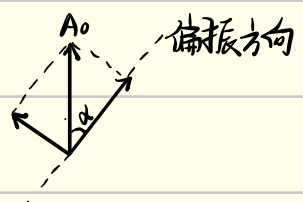
$E_x = E_0 \sin(kz - \omega t + \frac{\pi}{2})$
 $E_y = E_0 \sin(kz - \omega t)$

- Left(左)-handed (LCP)
- Electric field spirals CCW in space (at fixed time)

LCP = CCW

$E_x = E_0 \sin(kz - \omega t - \frac{\pi}{2})$
 $E_y = E_0 \sin(kz - \omega t)$

自然光在各个方向高速随机变化



$$A = A_0 \cos\alpha \quad \therefore A^2 = A_0^2 \cos^2\alpha$$

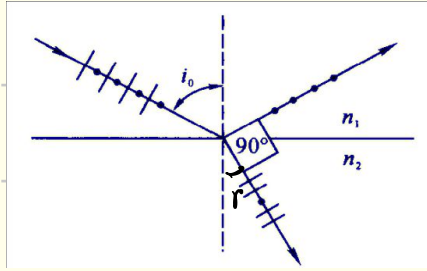
$$\frac{I}{I_0} = \frac{A^2}{A_0^2} = \cos^2\alpha$$

自然光通过第一个偏振片后

$$\Rightarrow I = I_0 \cos^2\alpha \quad \text{马吕斯定律} \quad \text{光强为原来的一半!}$$

布儒斯特定律：入射角某一*i*₀时，反射光成为线偏振光，振动方向垂直入射面，

折射光有全部平行振动成分，最少量的垂直振动成分。此时反射光与折射光垂直。



$$n_1 \sin i_0 = n_2 \sin(90^\circ - i_0) = n_2 \sin r \Rightarrow \frac{n_2}{n_1} = \tan i_0$$

$\tan i_0 = \underline{n_{21}}$ 媒质2对媒质1的相对折射率

*i*₀: 布儒斯特角, 起偏振角

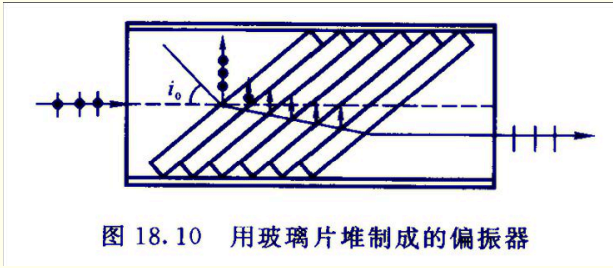


图 18.10 用玻璃片堆制成的偏振器

·光的双折射·

寻常光线: o光, 服从通常的折射定律; 非常光线, e光 传播方向不在入射面内

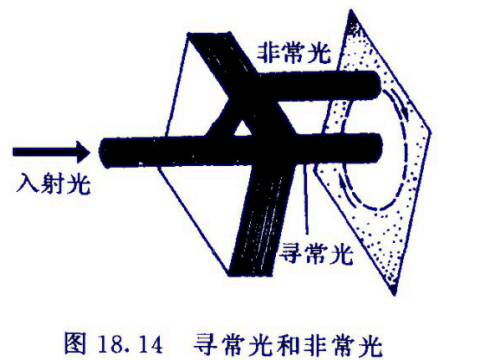


图 18.14 寻常光和非常光

都是线偏振光; 振动方向几乎互相垂直。

→ 垂直入射后, 转动晶体, e光转动。

光束沿某方向传播时不发生双折射, 这一方向称为晶体的光轴

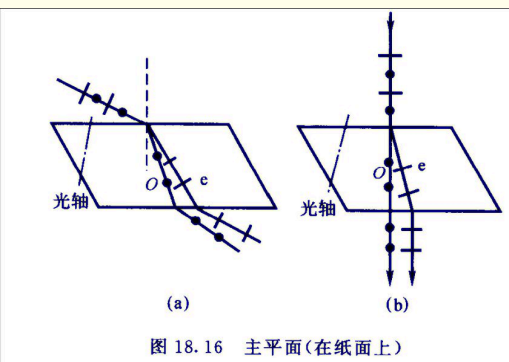
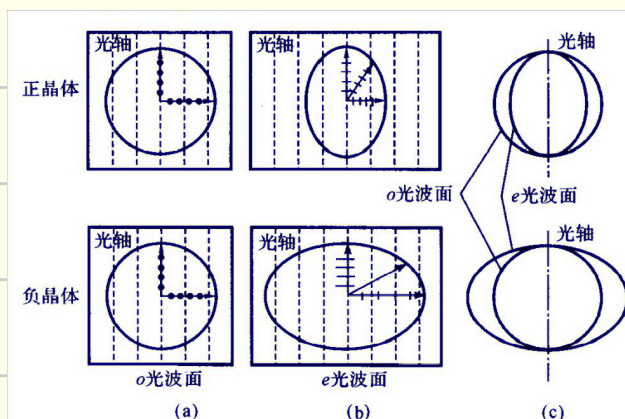


图 18.16 主平面(在纸面上)

原因: 晶体的规则结构 (各向异性)。

e光不遵从折射定律, *v* 也随方向而异



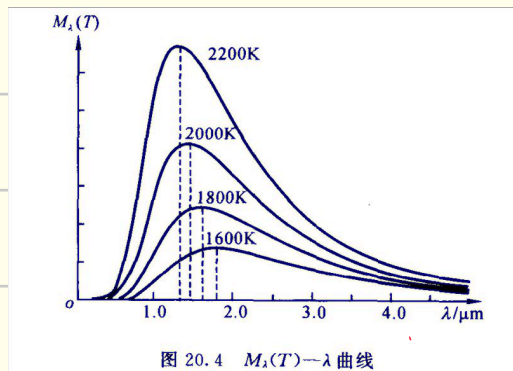
$n_e > n_o$

$n_e < n_o$

图 18.17 e光和o光的波面

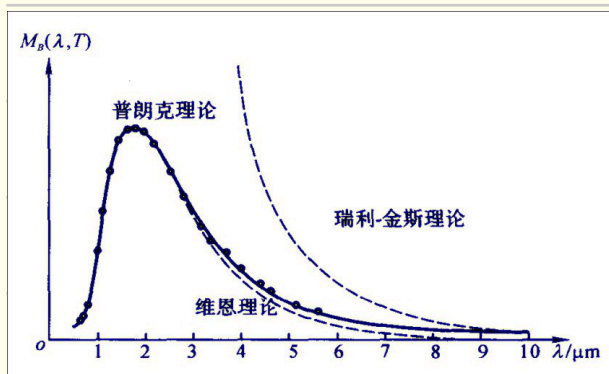
e光的主折射率: $n_e = \frac{c}{v_e}$ (e 沿垂直光轴方向的速度 v_e). n_o, n_e 为晶体主折射率.

黑体辐射



辐射出射度 $M_B(T) = \int_0^\infty M_{B\lambda}(T) d\lambda = \sigma T^4$ 斯特藩-玻尔兹曼定律

维恩位移定律: $T\lambda_m = b$.



普朗克公式: $R(\lambda, T) = \frac{2\pi hc^2}{\lambda^5 (e^{hc/\lambda kT} - 1)}$.

光电效应: $h\nu = K + \phi = \frac{1}{2}mv^2 + \phi$ → 逸出光电子的最大动能

截止频率 ν_0 $h\nu_0 = \phi$

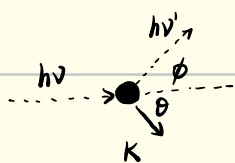
↓
单个光子能量 ↓ 金属逸出功.

$$K = h\nu - \phi = eV \Rightarrow V = \frac{h\nu}{e} - \frac{\phi}{e}$$

↑
遏止电压

康普顿效应:

$$\lambda f = \nu \star$$



$$E = h\nu = \hbar\omega; p = \frac{h}{\lambda} = \hbar k.$$

$$h = 2\pi\hbar$$

$$h\nu + m_0c^2 = h\nu' + mc^2 \Rightarrow \frac{hc}{\lambda_0} = \frac{hc}{\lambda'} + m_0c^2 \left(\frac{1}{\sqrt{1-(v/c)^2}} - 1 \right) \quad \text{--- ①}$$

$$\frac{h}{\lambda_0} = \frac{h}{\lambda'} \cos \phi + \frac{m_0 v}{\sqrt{1-(v/c)^2}} \cos \theta \quad \text{--- ②}$$

$$\frac{h}{\lambda'} \sin \phi = \frac{m_0 v}{\sqrt{1-(v/c)^2}} \sin \theta \quad \text{--- ③}$$

物质波(德布罗意波): $E = h\nu$, $p = \frac{h}{\lambda}$

$$\Rightarrow \Delta\lambda = \lambda' - \lambda_0 = \frac{h}{m_0c} (1 - \cos \phi).$$

不确定性原理: $\Delta x \cdot \Delta p \geq \hbar/2$. $\Delta E \Delta t \geq \hbar/2$. $\frac{\partial^2 f(x)}{\partial t^2} = -\omega^2 f(x)$ 时间上.

薛定谔方程: 从波动方程 $\frac{\partial^2 f(x)}{\partial x^2} = -k^2 f(x)$ 出发; $\frac{\partial^2 \psi}{\partial x^2} = -k^2 \psi$ 解形为 $\psi(x) = Ae^{i(kx - \omega t)}$ 空间上

$$p = \hbar k = k\hbar \Rightarrow k = \frac{p}{\hbar} \quad \therefore \frac{\partial^2 \psi}{\partial x^2} + \left(\frac{p}{\hbar}\right)^2 \psi = 0 \Rightarrow p^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

粒子总能量 $E = \frac{p^2}{2m} + V(x) \Rightarrow p^2 = 2mE - 2mV(x)$ 代入 $\Rightarrow 2m(E - V(x))\psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$

动能 势能

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x)\psi(x) = E\psi(x)$$

$$\omega = \frac{E}{\hbar}$$

一维定态方程.

含时部分: $\phi(t) = e^{-i\omega t} = e^{-i\frac{E}{\hbar}t}$ $\psi(x, t) = \psi(x) \cdot e^{-i\frac{E}{\hbar}t}$

$$\Rightarrow \frac{\partial \psi(x, t)}{\partial t} = \psi(x) \cdot \left(-i\frac{E}{\hbar} \cdot e^{-i\frac{E}{\hbar}t}\right) = -i\frac{E}{\hbar} \psi(x, t). \quad \textcircled{1}$$

代入一维定态: $\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)\right) \psi(x) \cdot e^{-i\frac{E}{\hbar}t} = E\psi(x) \cdot e^{-i\frac{E}{\hbar}t}$

$$\Rightarrow \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)\right) \psi(x, t) = E\psi(x, t)$$

由① $\Rightarrow E\psi(x, t) = -i\hbar \frac{\partial \psi(x, t)}{\partial t} = i\hbar \frac{\partial \psi(x, t)}{\partial t}$

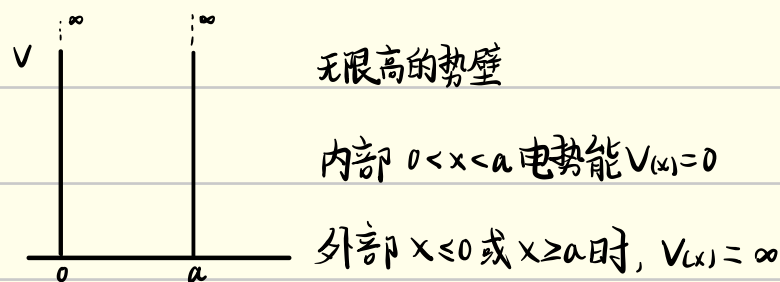
$$\Rightarrow \text{一维含时方程: } i\hbar \frac{\partial \psi(x, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t)\right) \psi(x, t)$$

三维含时: $i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}, t)\right) \psi(\vec{r}, t)$

三维定态: $\left(-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r})\right) \psi(\vec{r}) = E\psi(\vec{r})$

$\psi^2(x)$ 是粒子出现在 x 处的概率密度 $\psi^2(x) = \psi^*(x) \cdot \psi(x) \Rightarrow \iiint_V \psi^* \psi dV = 1$ 空间中粒子出现全概率为1.

9 一维无限深势阱中的粒子



由一维定态方程: $-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + V(x)\right) \psi(x) = E\psi(x) \Rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} = E\psi(x) \Rightarrow \frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2m}{\hbar^2} E\psi(x) = 0$

$$\Rightarrow \frac{\partial^2 \psi(x)}{\partial x^2} + k^2 \psi(x) = 0 \quad \therefore \psi(x) = A \sin kx + B \cos kx$$

$$(k = \frac{2mE}{\hbar^2})$$

在势阱壁: $\psi(0)=0, \psi(a)=0$ 此时 $\psi(x) = A \sin kx$

对于 $x=0$, 始终 $\psi(x)=0$; $\psi(a)=0 \Rightarrow \sin ka=0 \therefore ka=n\pi \Rightarrow k=\frac{n\pi}{a}, n=1, 2, 3, \dots$

即 $\psi(x) = A \sin \frac{n\pi x}{a} \therefore \int_0^a \psi^*(x) \psi(x) dx = 1 \Rightarrow A = \sqrt{\frac{2}{a}} \Rightarrow$ 粒子定态波函数 $\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x, n=1, 2, 3, \dots$

代入定态波函数到 $V(x)=0$ 的一维定态方程解得能量:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \quad n=1, 2, 3, \dots$$

$\Rightarrow$ 能量是量子化的

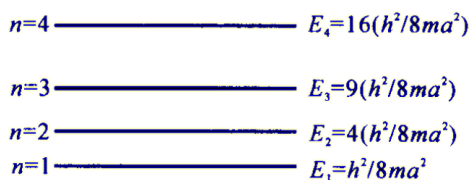


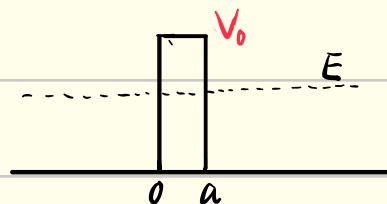
图 21.8 能级图

最小能量

$$E = \frac{\pi^2 \hbar^2}{2ma^2} \text{ 零点能}$$

由 $k = \frac{n\pi}{a} = \frac{2\pi}{\lambda_n} \Rightarrow$ 驻波波长: $\lambda_n = \frac{2a}{n} \quad n=1, 2, 3, \dots$

势垒 & 隧道效应



势能函数 $V(x) = \begin{cases} V_0, & 0 \leq x \leq a \\ 0, & x < 0, x > a \end{cases}$

一维定态: $E\psi(x) = (-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x))\psi(x)$

$$\Rightarrow \frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2m}{\hbar^2} (E - V(x)) \psi(x) = 0$$

$x < 0, x > a$ 时, $k_1 = \frac{1}{\hbar} \sqrt{2mE}$; $0 \leq x \leq a$ 时, $k_2 = \frac{1}{\hbar} \sqrt{2m(E - V_0)}$

$\therefore x < 0$ 时, $\psi_1(x) = A_1 e^{ik_1 x} + A_2 e^{-ik_1 x}$

$0 < x < a$ 时, $\psi_2(x) = B_1 e^{ik_2 x} + B_2 e^{-ik_2 x}$

$\Rightarrow$ 连续: $\begin{cases} \psi_1(0) = \psi_2(0) \\ \psi_2(a) = \psi_3(a) \end{cases}$

$x > a$ 时, $\psi_3(x) = C e^{ik_1 x}$

从 $x < 0$ 到 $x > a$ 的概率, 用透射率 T 表示, 定义为透射波与入射波强度之比

$$\Rightarrow T = \frac{|C|^2}{|A_1|^2} \propto e^{-\frac{2}{\hbar} \sqrt{2m(V_0 - E)} a} = e^{-2Kx}$$

$$K = \frac{1}{\hbar} \sqrt{2m(V_0 - E)}$$

6. 氢原子结构:

玻尔理论: 原子存在系列确定能量的稳定状态. (定态)

处于定态时, 电子稳定圆周: 角动量 $L = mvr = n\hbar$ $n=1, 2, 3, \dots$

轨道半径量子化: $\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2} = m \frac{v^2}{r} \Rightarrow r_n = n^2 \frac{\epsilon_0 h^2}{\pi m e^2}$ r_1 为玻尔半径 $r_n = n^2 r_1$

能量: $E_n = \frac{1}{2} m v_n^2 - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_n} = -\frac{1}{n^2} \left(\frac{m e^4}{8 \epsilon_0^2 h^2} \right)$ $n=1, 2, 3, \dots \Rightarrow E_n = \frac{E_1}{n^2}$

氢原子光谱: 高能级 i $\xrightarrow{\text{跃迁}}$ 低能级 f , 发射光子

$$E_i - E_f = h\nu = \frac{hc}{\lambda}$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ 里德伯常数.}$$